

# Evan Maurer

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## EDUCATION

**Missouri University of Science and Technology** — B.S. Computer Science May 2027  
Concentration: Artificial Intelligence & Machine Learning Rolla, MO

- Coursework: Data Structures & Algorithms, Statistics, Linear Algebra, Deep Learning, NLP, Statistical Machine Learning

## RESEARCH EXPERIENCE

**FarmVates Research Group** — *Undergraduate Researcher* Nov 2025 – Present

- Built domain-randomized sticky-trap synthetic image pipelines (Python, Blender/GLB, TripoSR) for cucurbit pest detection under scarce labeled real data.
- Trained and ablated YOLOv11 models (PyTorch / Ultralytics) with syn-pretrain → real-finetune curricula and **real-only validation** to measure sim-to-real transfer on Missouri S&T GPUs (V100 / A5000).
- Instrumented experiment logging, dataset prep, and domain-gap tooling; contributing to the SynthPest synthetic-insect data framework paper.

## ENGINEERING PROJECTS

**FarmVates Edge-to-Cloud Imaging Platform** — *Firmware & Systems* 2025 – 2026

- Shipped Pico 2 W (RP2350) camera firmware: ArduCAM OV2640 capture, Wi-Fi MQTT and cellular upload, OLED/watchdog/power-save duty cycles (C/C++, Pico SDK).
- Designed cellular OTA with A/B flash partitions, HTTPS manifest versioning, modem HTTPTOFS download, SHA256 verify, and inactive-slot UF2 programming.
- Implemented on-device scene-change gate (160×120 → 96×96 Y, SSIM/MAE ablations) to skip redundant UXGA + radio; field-scored 245 uploaded frames with offline ROC tools.
- Contributed to cloud path: ZeroMQ/Redis message worker, PyTorch dark-image CNN gate before YOLO inference, Docker/Jenkins services, Flask/Mongo APIs, SwiftUI iOS client.

**MontageAI** — *Independent AI Desktop App (in progress)* 2026

- Built a working Electron + React + FastAPI montage editor: project shell, media analysis pipeline, and editable AI plan objects (scoring, highlights, music sync, pacing) — not a one-click black-box render.
- Added Albion-focused detectors (combat timelines, abilities, bomb fusion, searchable highlights); full export/polish roadmap remains in progress.

**Simply Natural** — *Website Designer & Maintainer* Aug 2021 – Present

- Maintain production WordPress site integrated with LightSpeed POS for live inventory.

## SKILLS

**Languages:** Python, C/C++, TypeScript, Swift, SQL

**AI / CV:** PyTorch, Ultralytics YOLO, OpenCV, scikit-learn, CUDA, synthetic data / domain randomization

**Systems:** Pico SDK, MQTT, cellular modems, OTA, Docker, Redis, ZeroMQ, Flask, FastAPI, MongoDB, Jenkins, S3

**Product:** Electron, React, SwiftUI, Git/Linux